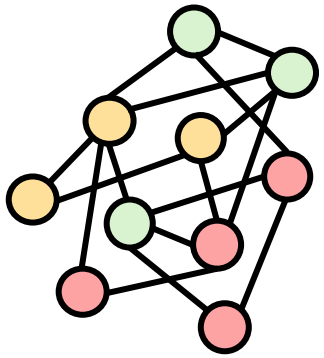
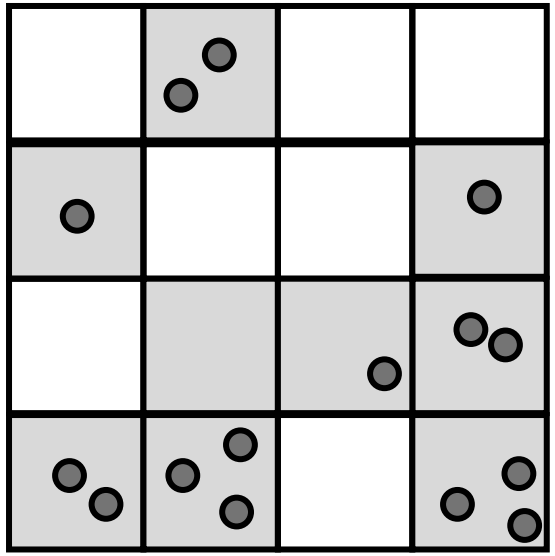
Initialization (pre-computed steps)



High-dimensional data

PCA

N-dimensional cube (2-D example)



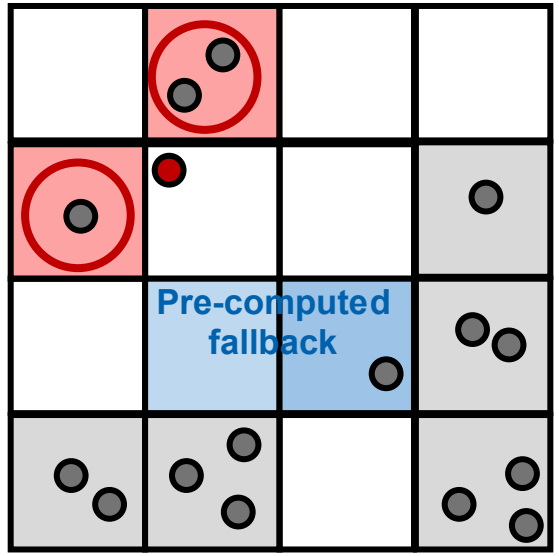
BFS maps empty cells to neighboring occupied cells

= empty cell

= occupied cell

Similarity search

PCA



- Check primary cell and n_probe-1 closest neighbors
- If all are empty: fallback to pre-computed nearest neighbor
- Rank candidates in the native space